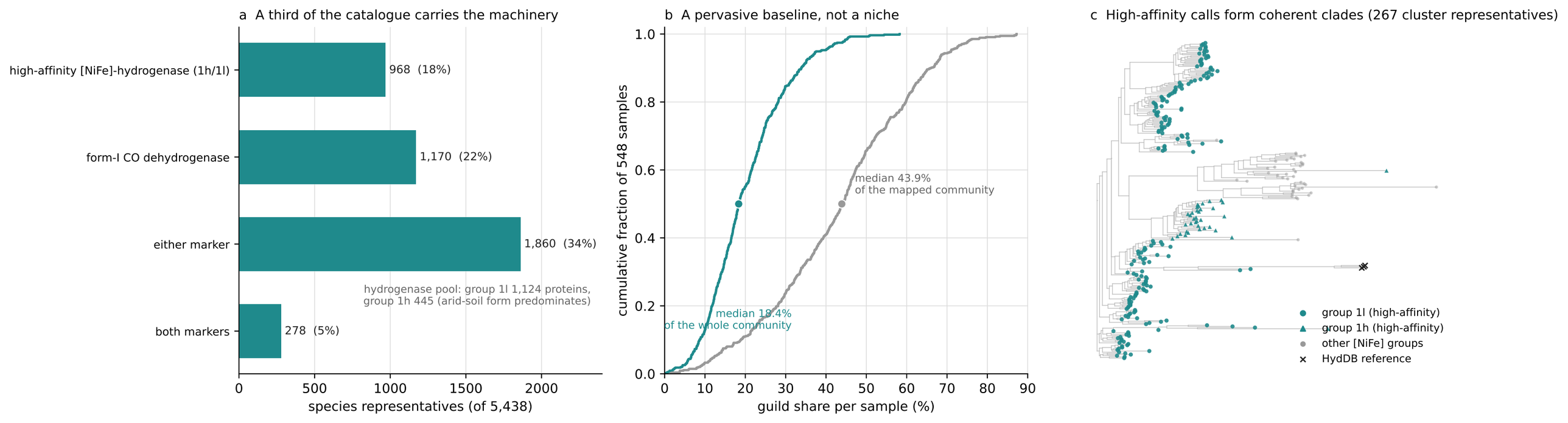

a A third of the catalogue carries the machinery
high-affinity [NiFe]-hydrogenase (1h/1l)
968 (18%)
form-I CO dehydrogenase
1,170 (22%)
either marker
1,860 (34%)
hydrogenase pool: group 1l 1,124 proteins,
group 1h 445 (arid-soil form predominates)
both markers
278 (5%)
0
500
1000
1500
2000
species representatives (of 5,438)
b A pervasive baseline, not a niche
cumulative fraction of 548 samples
0.0
0.2
0.4
0.6
0.8
1.0
median 18.4%
of the whole community
median 43.9%
of the mapped community
0
10
20
30
40
50
60
70
80
90
guild share per sample (%)
c High-affinity calls form coherent clades (267 cluster representatives)
group 1l (high-affinity)
group 1h (high-affinity)
other [NiFe] groups
HydDB reference